

Agent-Level Evaluation Report

Data Analyst Digital Twin — 2026-03-17 12:48

Methodology

This report follows established agent evaluation frameworks: **G-Eval** (Liu et al., 2023) for criteria-based rubric scoring, **MT-Bench** (Zheng et al., 2023) for multi-turn task completion, and **Behavioral Contract Testing** for agent invariants. Each scenario is scored on a 1–5 rubric across multiple dimensions.

Summary Metrics

Metric	Value
Total Scenarios	31
Passed	27
Failed	4
Pass Rate	87.1%
Total Score	231 / 245
Overall Score	94.3%

Scores by Category

Category	Scenarios	Passed	Score	%
Cleaning	2	2	20/20	100%
Dataset Resolution	4	4	20/20	100%
EDA	3	2	37/40	92%
Error Handling	4	4	20/20	100%
Full Pipeline	1	1	15/15	100%
SQL	2	1	20/25	80%
Statistics	2	1	22/25	88%
Tool Selection	11	11	55/55	100%
Visualization	2	1	22/25	88%

Detailed Scenario Results

ID	Scenario	Category	Score	Pass?
eda-1	EDA Profile	EDA	15/15	PASS
eda-2	EDA Correlations	EDA	12/15	FAIL

eda-3	Data Quality Check	EDA	10/10	PASS
viz-1	Bar Chart Generation	Visualization	12/15	FAIL
viz-2	Scatter Plot Generation	Visualization	10/10	PASS
sql-1	SQL Schema Inspection	SQL	7/10	FAIL
sql-2	SQL Query Execution	SQL	13/15	PASS
stats-1	Statistical Assumptions Check	Statistics	12/15	FAIL
stats-2	Hypothesis Test (auto-select)	Statistics	10/10	PASS
clean-1	Structural Cleaning	Cleaning	10/10	PASS
clean-2	Data Validation	Cleaning	10/10	PASS
route-eda_profile	Route: 'profile my data'	Tool Selection	5/5	PASS
route-eda_data_quality	Route: 'check data quality'	Tool Selection	5/5	PASS
route-eda_describe	Route: 'describe the columns'	Tool Selection	5/5	PASS
route-eda_correlations	Route: 'show correlations'	Tool Selection	5/5	PASS
route-sql_schema	Route: 'show the schema'	Tool Selection	5/5	PASS
route-clean_validate	Route: 'validate the data'	Tool Selection	5/5	PASS
route-focused-write-sql	Route: 'write a SQL query for top 1'	Tool Selection	5/5	PASS
route-focused-clean	Route: 'clean the data'	Tool Selection	5/5	PASS
route-focused-create-bar-chart	Route: 'create a bar chart'	Tool Selection	5/5	PASS
route-focused-run-statistic	Route: 'run a statistical test'	Tool Selection	5/5	PASS
route-full-analyze-everything	Route: 'analyze everything in the d'	Tool Selection	5/5	PASS
resolve-1	Resolve by exact ID	Dataset Resolution	5/5	PASS
resolve-2	Resolve by filename	Dataset Resolution	5/5	PASS
resolve-3	Resolve by dataset_name param	Dataset Resolution	5/5	PASS
resolve-4	Resolve empty ID (single dataset fa	Dataset Resolution	5/5	PASS
err-1	Nonexistent dataset (graceful failu	Error Handling	5/5	PASS
err-2	Unknown tool name (graceful failure	Error Handling	5/5	PASS
err-3	Missing required param (no x/y)	Error Handling	5/5	PASS
err-4	No data loaded (supervisor help mes	Error Handling	5/5	PASS
pipe-1	Full Direct Pipeline (profile)	Full Pipeline	15/15	PASS

Scoring Rubric (G-Eval 1–5 Scale)

- 1 — Failure — no result, crash, or completely wrong output
- 2 — Poor — partial result but critical issues (wrong type, missing key data)
- 3 — Adequate — result produced but with minor issues (imprecise, extra fields)
- 4 — Good — correct result, properly structured, minor polish needed
- 5 — Excellent — correct, well-structured, informative summary, no issues

Evaluation Frameworks Used

- **G-Eval** (Liu et al., 2023) — Criteria-based rubric scoring with evidence
- **MT-Bench** (Zheng et al., 2023) — Multi-step task completion assessment
- **Behavioral Contract Testing** — Agent invariants that must hold regardless of LLM variance
- **Adversarial Testing** — Intentionally invalid inputs to verify graceful degradation